Basic Principles of Medicine 2
Module : Endocrine and reproductive Systems
ER DLA

DLA Title: Developmental Genetics

Hox genes & Achondroplasia

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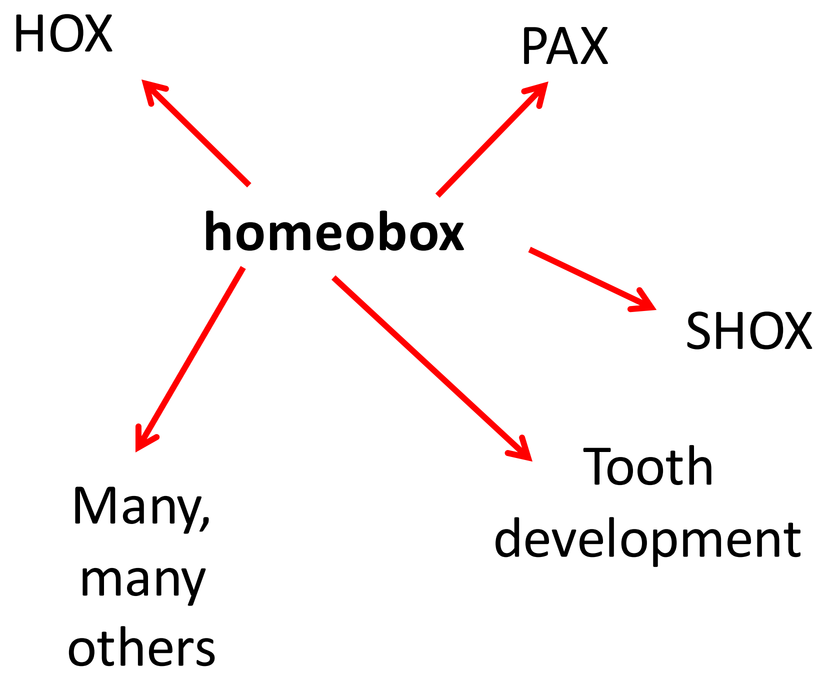
Learning Objectives:

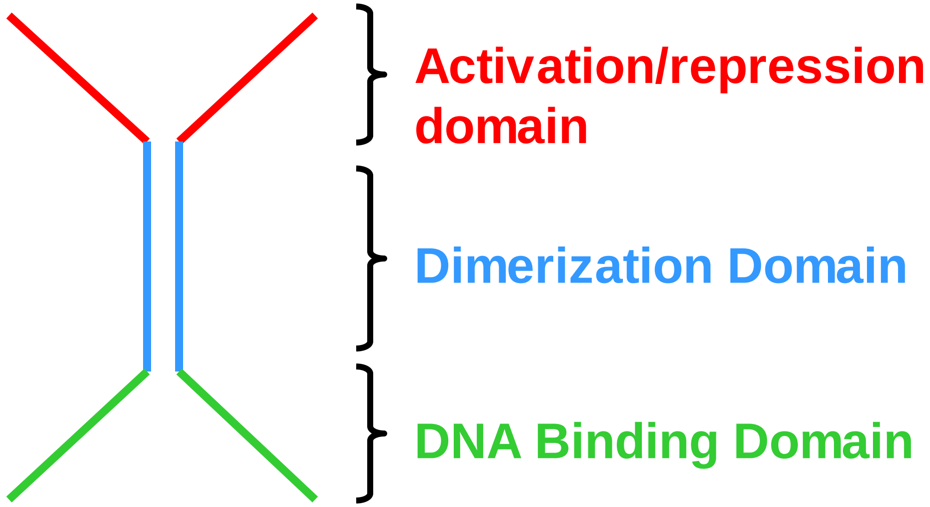
SOM.MK.III.BPM2.4.ER.3.GNET.1005	Describe the HOX gene
SOM.MK.III.BPM2.4.ER.3.GNET.1006	Illustrate the function of the HOX genes
SOM.MK.III.BPM2.4.ER.3.GNET.1007	Discuss disorders associated with HOX genes
SOM.MK.III.BPM2.4.ER.3.GNET.1008	Describe the general features of achondroplasia
SOM.MK.III.BPM2.4.ER.3.GNET.1009	Discuss the inheritance pattern of the achondroplasia
SOM.MK.III.BPM2.4.ER.3.GNET.1010	Explain the possible outcomes to the offspring of two parents who both have achondroplasia
SOM.MK.III.BPM2.4.ER.3.GNET.1011	Analyze the effect of the FGFR3 G380R mutation and explain why the disorder can be classified as autosomal dominant to gain of function

Hox genes:

- Subset of the Homeobox gene family.
- Homeobox: DNA binding domain of the protein = homeodomain
- Many homeobox genes: About 235 functional genes and many pseudogenes in humans

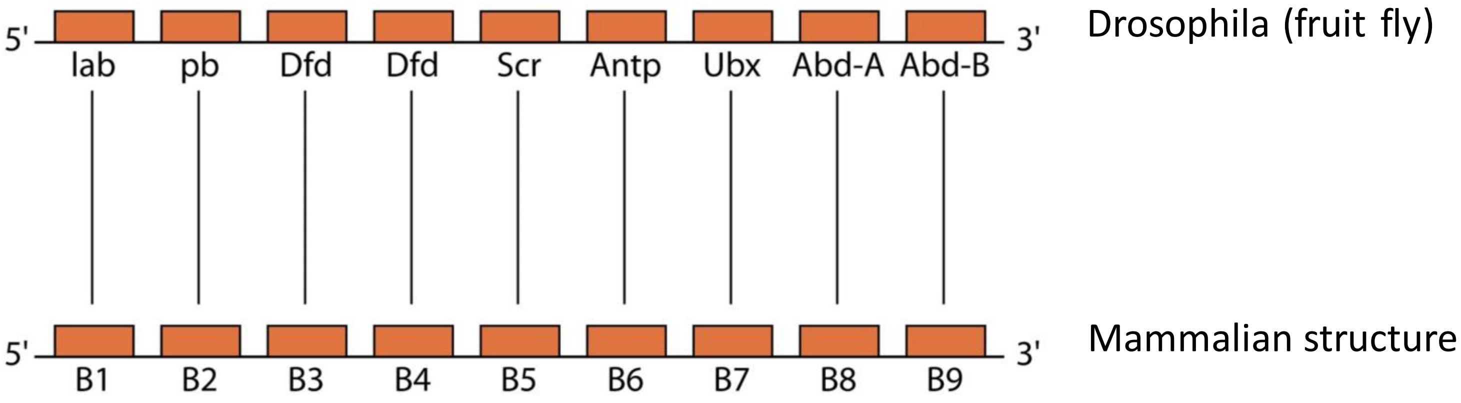


- Hox proteins are **transcription factors**
- Form dimers when bound to DNA.
- Global regulators of development.
- HOX genes arranged in clusters: collinear expression



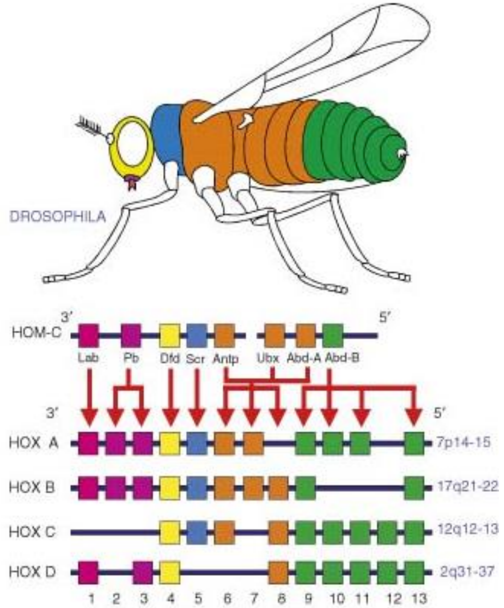
Hox gene function:

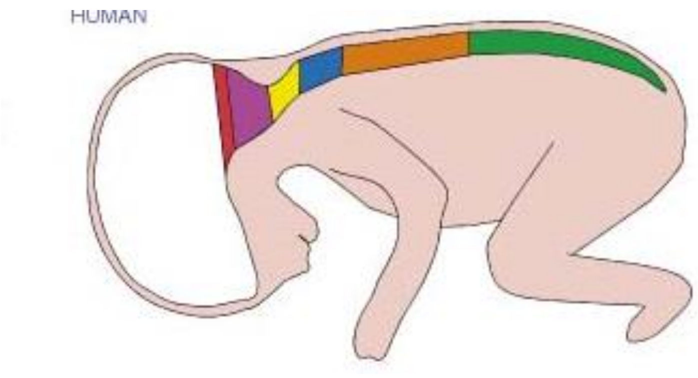
- Collinear expression of Hox genes essential for development.
- As a body segment (or body part) needs instruction to develop, the next HOX gene is expressed.
- Sequential expression during development controls body part formation.



Mammals have four of these repetitive structures (on different chromosomes)

Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.





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Disorders associated with Hox gene dysfunction:

- Most severe phenotypes are not seen:
  - ? Incompatible with life
- Diverse disorders associated with mutation: >180 different conditions
  - An extra cervical rib
  - Polydactyly
  - Increased risk for cancer
  - Eye development
  - Ear development
  - Short Stature
  - Head development anomaly
  - Face development anomaly
  - Tooth development anomaly

Synpolydactyly:
HoxD13 mutation

Image reproduced from Essential Genetics – Klug et al.

https://commons.wikimedia.org/wiki/File:Waardenburg_syndrome_type_1_(3).png

https://en.wikipedia.org/wiki/Waardenburg_syndrome

Waardenburg syndrome:
Pax3 or Pax6 mutation

SOM.MK.III.BPM2.4.ER.3.GNET.1007

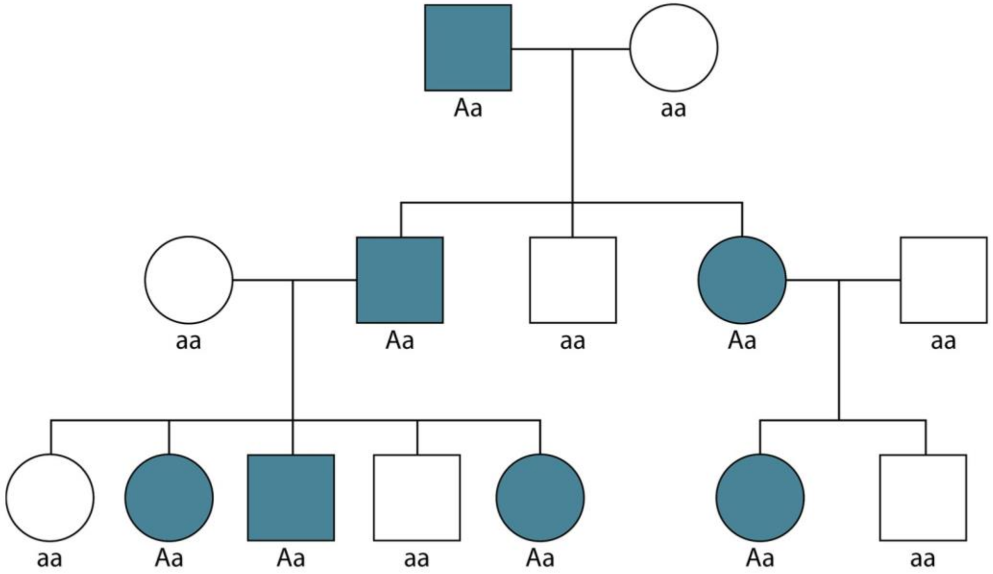
Achondroplasia:

- Form of dwarfism
- General features:
  - Short arms and legs with a normal torso size
  - Reduced stature (height)
  - Facial features: Large head size and flattening of bridge of the nose
  - Shortened fingers
  - Trident hand (separation between middle and adjacent fingers)

https://www.clinicalultrasound.net/Education/Obstetrics/Fetal%20Limbs%20april%2021.pdf

Genetics of Achondroplasia:

- Autosomal dominant.
- Gain of function mutation in the FGFR3 gene.
- Full penetrance between generations.
- If one parent is affected, risk to offspring = 50%
- If both parents are affected, risk = 75% chance of being affected (50% heterozygous + 25% homozygous) and only a 25% chance of a normal child.
- Homozygous for mutated allele = lethal



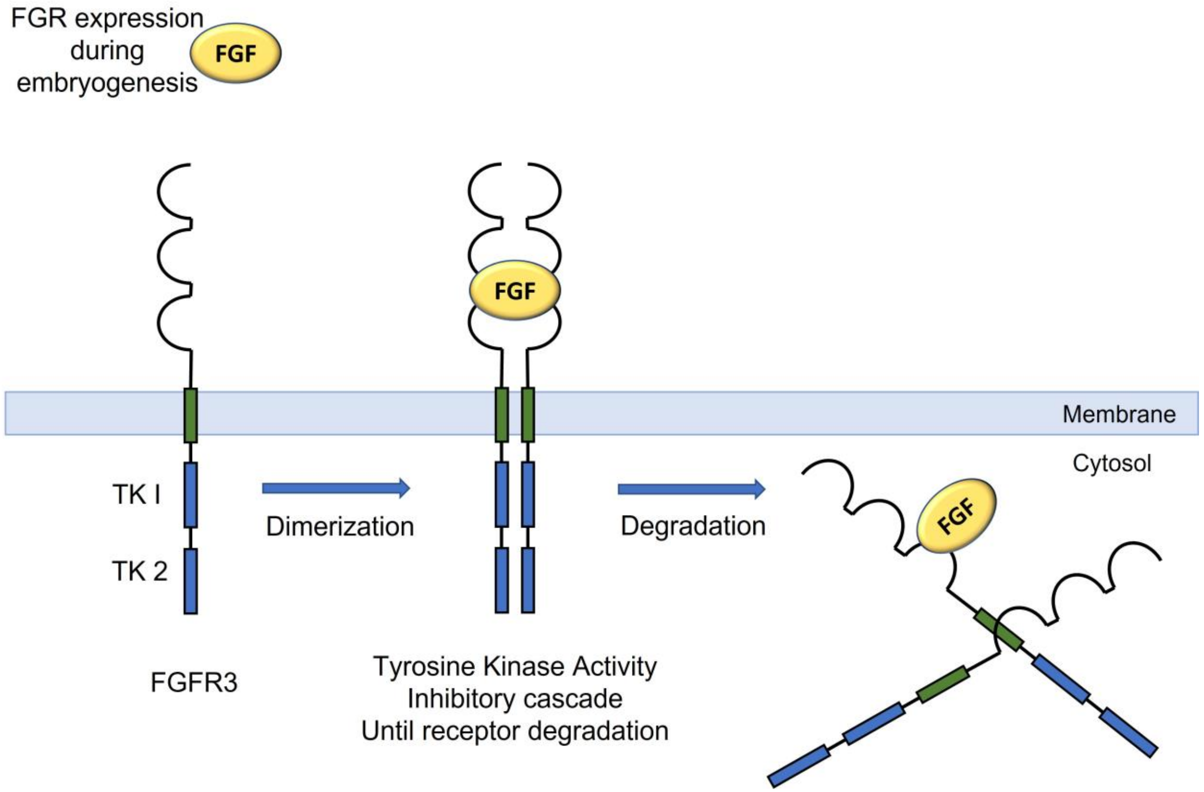
Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.

Function of FGFR3

- Transmembrane receptor tyrosine kinase

- Ligand binding outside the cell leads to intracellular responses.

- Involved in the promoting cartilage differentiation into bone - Ossification

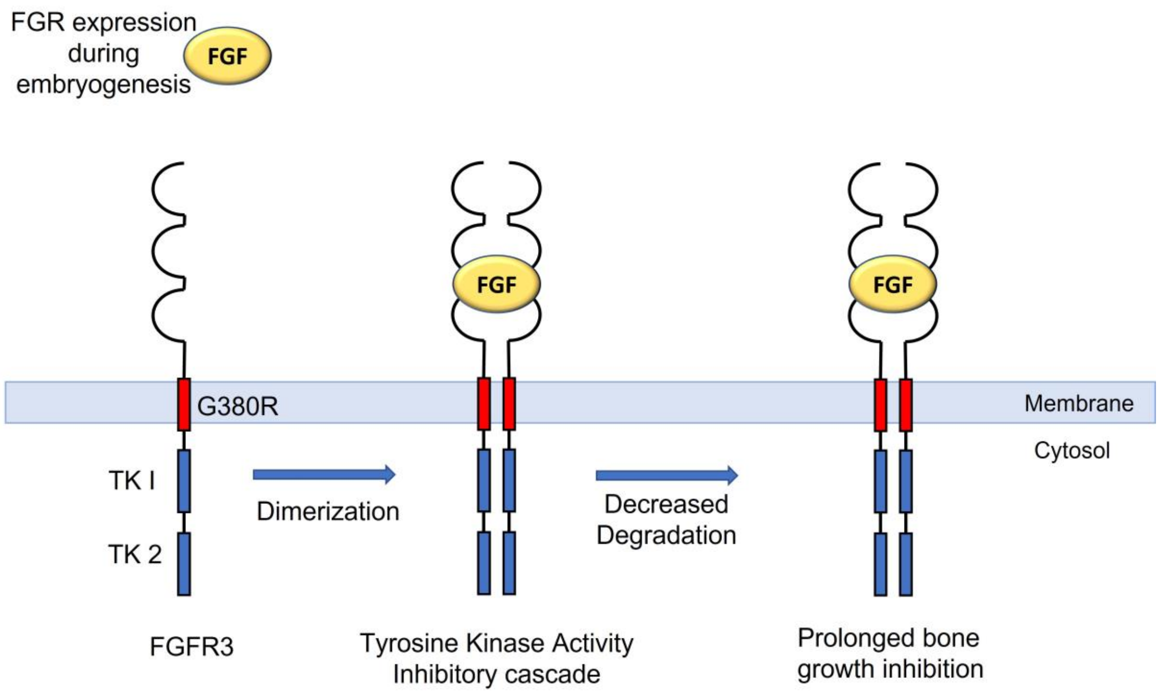


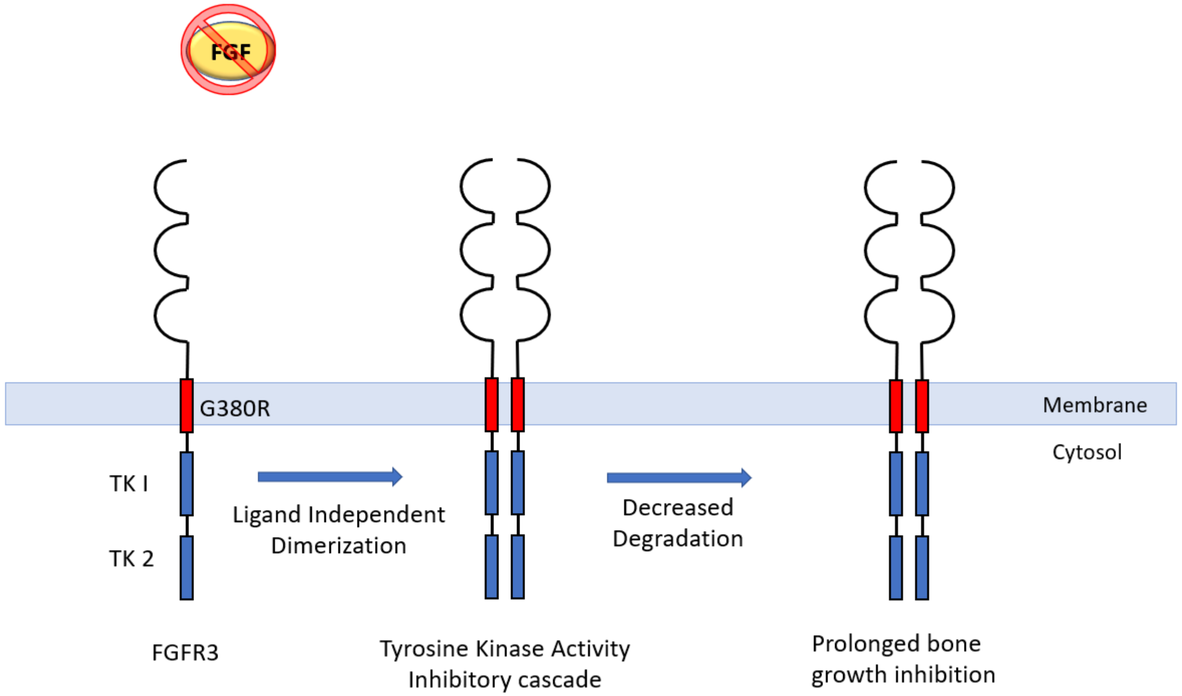
Function of FGFR3

- G380R mutation causes **unregulated activity** = growth plate prematurely converted to bone.

- "**Gain of function** mutations"

- Other mutations on *FGFR3* lead to skeletal dysplasia disorders; Very rare





Summary and Genetic Principles Illustrated:

Hox genes:

- Global regulators of development
- Transcription factors
- Clustered arrangement with collinear expression
- Sequential expression during development controls body part formation

Achondroplasia:

- Autosomal Dominant Inheritance
- Fully penetrant
- Gain of function mechanism
- Almost all known cases due to the same mutation- potential idea of a mutation “hot-spot” in the *FGFR3* gene

Thank you for listening!